

GROUP 12 - WEEKLY SUMMARIES

Milestone 1 – Weeks 1 & 2

Week 1 we established the project foundation, covering problem definition, initial data acquisition, and exploratory analysis. We defined our central research question: How do CH₄ and N₂O emissions vary across countries, and how effectively can emission reduction status be predicted using per-capita emissions and geographic classification? We sourced a real-world UNSDemissions dataset from Kaggle.

Week 2 covered concepts of Data Quality Assessment, Missing Value Identification, and Renaming for Clarity. We conducted systematic missing value assessment and discovering significant gaps. We applied `rename()` to convert confusing column names (`X..change.since.1990`) into human-readable ones (`CH4.change.since.1990`), establishing a best practice for code maintainability.

Milestone 2 – Weeks 3 & 4

Milestone 2 focused on cleaning, transformation, and univariate analysis to prepare our data for deeper investigation.

Week 3 covered concepts of Type Conversion, Duplicate Handling, and Systematic Missing Value Imputation. We learned to use `suppressWarnings(as.double())` to safely convert character columns (like `N2O.emissions`) to numeric, handling coercion errors gracefully. We then identified and removed duplicate rows using `unique()` and `duplicated()`.

Week 4 covered concepts of Data Transformation, Skewness Correction, and Basic Feature Engineering. We created a `Total_Emissions` feature and began categorical classification using `case_when()`, transforming raw numbers into actionable policy insights.

Milestone 3 – Weeks 5 & 6

Milestone 3 Visualization & Exploratory Analysis

Week 5 Charts & Graphs Created: **A Histograms for CH₄ and N₂O Emissions** - we created separate histograms using `geom_histogram()` to visualize the frequency distribution of emission values across all countries. **Scatter Plots for CH₄ vs. N₂O Relationship** - we created scatter plots using `geom_point()` to visualize the bivariate relationship between CH₄ and N₂O emissions and **A top 10 Emitters Bar Chart**.

Week 6 covered concepts of Comparative Visualization Design and Feature Engineering for Modeling. We learned to compare bar charts vs. boxplots, articulating why boxplots on log scales are superior for skewed emission data, as they correctly display medians and highlight extreme outliers without distortion.

Milestone 4 – Weeks 7 & 8

Milestone 4 focused exclusively on **Statistical Inference & Analytical Modeling**

Week 7 covered concepts of Hypothesis Testing (t-test and Chi-Square) for Categorical and Continuous Comparisons. We performed **independent two-sample t-tests** to compare mean N₂O emissions between "Safe" and "Dangerous" methane-emitting countries. The significant result ($p < 0.001$) statistically confirmed that dangerous methane emitters also have significantly higher N₂O emissions.

Week 8 covered concepts of Correlation Analysis, Linear Regression, and Interpretation of Uncertainty. We calculated and interpreted **Pearson correlation coefficients** between all numeric variables, creating a correlation matrix to identify multicollinearity concerns.

Milestone 5 – Weeks 9 & 10

Milestone 5: Interactive Visual Analytics System.

Week 9 covered concepts of Interactive Dashboard Architecture, Filter Controls, and User-Centered Design Principles. We designed a **Shiny dashboard** with a modular UI using `dashboardPage`, `sidebarMenu`, and `tabItems`. We implemented **global filter controls** (region selector) that allow users to dynamically subset data across all visualizations simultaneously, learning the concept of reactive data filtering.

Week 10 covered concepts of KPI Tracking, Performance Metrics, and Real-Time Interactive Visualizations. We implemented **KPI tracking** using `valueBox` components that display real-time metrics: total countries analyzed, average CH₄ emissions, average N₂O emissions, and percentage of dangerous emitters. These update dynamically based on filter selections.

Milestone 6 – Weeks 11 & 12

Milestone 6: Research Contribution & Advanced Analytics

Week 11 covered concepts of Ensemble Model Comparison, Novel Weighted Priority Scoring, and Advanced Risk Classification. We implemented **advanced model evaluation** using AUC-ROC curves, precision-recall analysis, and feature importance extraction. Random Forest achieved 94.4% accuracy and 0.965 AUC, demonstrating that ensemble methods can effectively predict dangerous emission status from limited features.

Week 12 covered concepts of Interactive Simulation, What-If Analysis Tools, and Downloadable Decision Support Outputs. We designed a **"What-If" Simulation** tool with interactive sliders for CH₄ and N₂O emissions, allowing users to see in real-time how policy changes could move a country from "Dangerous" to "Safe." We added a downloadable CSV export for the priority list

and created an explanatory UI component that dynamically explains why the top-priority country requires immediate intervention, transforming the application from a static report into an active decision-support system.